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# Review of Empirical Studies on IPO Activity and Pricing Behavior in Korea

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## Abstract

This paper reviews the hypotheses and empirical evidence that have been proposed to explain IPO activities and pricing behavior in Korea. The paper particularly reviews empirical evidence and explanations related to the three patterns associated with Korean IPOs: underpricing, hot issue markets, and long-run underperformance. Explanations for underpricing can be grouped under two broad categories: deliberate underpricing and behavioral approaches. The first category can be divided into two subcategories: asymmetric information and price stabilization. While the first subcategory attempts to explain underpricing in the efficient market framework and consider it as a part of the costs associated with IPOs, the last subcategory turns to one of the services that underwriters use to provide price stabilization. While the hypotheses based on asymmetric information are well established and supported by the literature, the significant variation in the degree of underpricing over time together with the poor long-term performance makes many researchers doubtful whether information-based theories could be enough to explain the large initial returns. Some researchers thus turn to behavioral explanations which put more emphasis on the effect of investors' irrational behavior or sentiment on stock prices. Evidence suggests that no hypothesis can dominantly explain large initial returns. Thus the hypotheses reviewed here are not mutually exclusive and a given hypothesis can be more relevant to some IPOs than to others.

## Keywords

IPO Pricing, Underpricing, Price Stabilization, Hot Issue Markets, Long-Run Performance

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## I. Introduction

Initial public offering of shares (IPO) provides access for a privately held firm to raise equity capital in a public market and may lower the cost of capital needed for the firm's operations and investments. It also provides an opportunity for current shareholders to realize their capital gains and diversify their investments. But IPOs can be quite risky to each of the three major parties involved: issuers, investment bankers, and investors. The big risk stems from the pricing of IPO shares which is complicated by the fact that the valuation of shares is based on relatively incomplete and imperfect information since issuers tend to be young, private firms.

If the price is set too high, then the investor will earn an inferior return and consequently might avoid the IPO market. If it is priced too low, the issuer does not gain the full advantage of its ability to raise capital and will be unwilling to participate in the IPO market.

The empirical evidence on the pricing of IPOs provides a puzzle to those who otherwise believe in efficient markets. Many empirical studies have shown that when firms go public, their shares tend to be significantly underpriced, for which the market price jumps substantially on the initial day of trading. That is, the initial returns, estimated as the percentage difference between the offer price and the initial market price, are positive. Numerous studies document that this phenomenon exists in every nation with a stock market, although the extent of initial returns varies from country to country. The Korean IPO market is no exception to the phenomenon. The average initial return of IPOs is 57.6% over the period 2000~2007 (Lee and Kim 2009). It is markedly larger than the extent of initial returns in the U.S., which averages 25.68% over the period 2000~2007.<sup>1)</sup>

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1) The average initial return is calculated by using IPO data available from the homepage of Jay R. Ritter, a professor of University of Florida.

Given the level of competition and information efficiency in financial markets, this is a surprising result and is considered anomalous to the efficient market hypothesis. A number of hypotheses have been offered to explain this phenomenon in the Korean IPO market, focusing on various aspects of the relations among underwriters, issuers, and investors. In general, these hypotheses can be grouped under two broad categories: deliberate underpricing and behavioral approaches. The first category can be further divided into two subcategories: asymmetric information and price stabilization. While the first subcategory attempts to explain underpricing in the framework of the efficient market and consider it as a part of the necessary costs associated with initial public offerings, the last subcategory turns to one of services that underwriters provide in connection with IPOs -the price support practice which is designed to reduce price drops in the aftermarket. Behavioral approaches regard initial returns as fads or bubbles caused by irrational investors who bid up the price of IPO shares beyond true value.

Empirical evidence documented in the literature suggests that each of the hypotheses alone cannot fully account for the large initial returns observed in the Korean IPO market. The hypotheses reviewed in this paper thus are not mutually exclusive and a given hypothesis can be more relevant to some IPOs than to others.

The purpose of this paper is to review the hypotheses and empirical evidence that have been proposed to explain IPO activities and pricing behavior in the Korean market. The paper particularly reviews the three patterns associated with IPOs: large initial returns, hot issue markets, and long-run underperformance. The IPO literature in Korea is now fairly mature and has accumulated a large body of evidence, which is enough to cast light on the Korean IPO market. Because of limited space, however, this paper sets its main focus on papers presented in academic finance journals such as the Asia-Pacific

Journal of Financial Studies, The Korean Journal of Finance (Asian Review of Financial Research), The Korean Journal of Financial Management, and The Korean Journal of Financial Studies.

The remainder of this paper is organized as follows. First, evidence regarding IPO underpricing is presented. Second, the principal hypotheses proposed to explain IPO underpricing are discussed under the two subcategories mentioned above: information asymmetry and price stabilization. Third, the cyclical pattern of IPO activities (i.e., hot and cold issue markets) and the aftermarket performance of IPOs in the long run are presented and discussed in conjunction with behavioral explanations. Finally, concluding remarks are provided.

## **II . Evidence of IPO Underpricing in the Korean IPO Market**

The well-known phenomenon associated with going public is one of large initial returns accruing to investors in the secondary market. These initial returns are considered evidence of the underpricing of IPOs. <Table 1> summarizes the evidence of underpricing in the Korean IPO markets documented by empirical studies since Kang (1990). As shown in <Table 1>, there have been in general two measures used to show the extent of underpricing: initial returns and market-adjusted initial returns. Initial returns are estimated as the percentage difference between the offer price and the market price on the first day of trading. Market-adjusted abnormal returns are initial returns adjusted by market returns which are the percentage difference between the market index on the offer day and the market index on the first trading day. The difference between the two measures appears to be minimal according to several studies including Kang (1990), Jung (1992), Joo (1995), Choi (1999), Choi and Huh (2000), Choi (2000), and Lee and Joh (2007). For instance,

&lt;Table 1&gt; Evidence of Underpricing in the Korean IPO market

Evidence is collected from the academic finance journals such as the Asia–Pacific Journal of Financial Studies, The Korean Journal of Finance (Asian Review of Financial Research), The Korean Journal of Financial Management, and The Korean Journal of Financial Studies.

| Average under-pricing | Underpricing measures <sup>1</sup> | Sample size | Sample firms | Sample period   | Authors                  |
|-----------------------|------------------------------------|-------------|--------------|-----------------|--------------------------|
| 80.84%                | Market adjusted abnormal returns   | 126         | KSE firms    | 1988~1989       | Kang (1991)              |
| 99.80%                | Initial returns                    | 153         | KSE firms    | 1988~1989       | Kang (1990)              |
| 97.20%                | Market adjusted abnormal returns   | 153         | KSE firms    | 1988~1989       | Kang (1990)              |
| 70.00%                | Initial returns                    | 361         | KSE firms    | 1980.09~1991.04 | Jung (1992)              |
| 68.70%                | Market adjusted abnormal returns   | 361         | KSE firms    | 1980.09~1991.04 | Jung (1992)              |
| 54.21%                | Initial returns                    | 108         | KSE firms    | 1989~1994       | Joo (1995)               |
| 54.56%                | Market adjusted abnormal returns   | 108         | KSE firms    | 1989~1994       | Joo (1995)               |
| 30.62%                | Initial returns*                   | 61          | KSE firms    | 1990~1992       | Jang and Woo (1997)      |
| 75.45%                | Market adjusted abnormal returns   | 342         | KSE firms    | 1988~1994       | Lim (1997)               |
| 77.96%                | Initial returns                    | 62          | KSE firms    | 1994.01~1996.09 | Choi (1999)              |
| 78.70%                | Market adjusted abnormal returns   | 62          | KSE firms    | 1994.01~1996.09 | Choi (1999)              |
| 69.55%                | Initial returns                    | 107         | KSE firms    | 1992.01~1996.10 | Choi and Huh (2000)      |
| 70.21%                | Market adjusted abnormal returns   | 107         | KSE firms    | 1992.01~1996.10 | Choi and Huh (2000)      |
| 62.74%                | Initial returns                    | 141         | KSE firms    | 1992~1998       | Choi (2000)              |
| 64.11%                | Market adjusted abnormal returns   | 141         | KSE firms    | 1992~1998       | Choi (2000)              |
| 19.01%                | Market adjusted abnormal returns   | 130         | KOSDAQ firms | 1999.01~2000.12 | Huh, Yoon and Lee (2002) |
| 58.29%                | Initial returns                    | 125         | KSE firms    | 1992~1998       | Choi (2003)              |
| 14.98%                | Market adjusted abnormal returns   | 161         | KOSDAQ firms | 1997~2000       | Lee and Yi (2003)        |
| 56.66%                | Market adjusted abnormal returns   | 39          | KSE firms    | 1997~2000       | Lee and Yi (2003)        |
| 31.69%                | Market adjusted abnormal returns   | 360         | KOSDAQ firms | 2000.09~2003.12 | Choi (2005)              |

&lt;Table 1&gt; Continued

| Average underpricing | Underpricing measures <sup>1</sup>          | Sample size | Sample firms           | Sample period   | Authors                    |
|----------------------|---------------------------------------------|-------------|------------------------|-----------------|----------------------------|
| 57.00%               | Initial returns                             | 453         | KOSDAQ firms           | 2000.07–2005.05 | Choi and Jun (2006)        |
| 74.90% <sup>2</sup>  | Market adjusted cumulative abnormal returns | 256         | KOSDAQ firms           | 1999.11–2001.01 | Kim and Lee (2006)         |
| 61.00%               | Market adjusted abnormal returns            | 343         | KOSDAQ firms           | 2001.01–2003.09 | Park, Lim, and Sung (2006) |
| 61.10%               | Market adjusted abnormal returns            | 363         | KOSDAQ firms           | 2001.01–2003.12 | Park, Kim, and Sung (2007) |
| 45.50%               | Market adjusted abnormal returns            | 174         | KSE (20), KOSDAQ (154) | 2003.09–2006.06 | Lee and Joh (2007)         |
| 45.60%               | Initial returns                             | 174         | KSE (20), KOSDAQ (154) | 2003.09–2006.06 | Lee and Joh (2007)         |
| 34.70%               | Market adjusted abnormal returns            | 50          | KSE firms              | 1999.01–2006.12 | Park and Shin (2007)       |
| 46.20%               | Market adjusted abnormal returns            | 819         | KOSDAQ firms           | 1999.01–2006.12 | Park and Shin (2007)       |
| 45.79%               | Initial returns                             | 33          | KSE firms              | 2000.02–2007.07 | Lee and Kim (2009)         |
| 58.29%               | Initial returns                             | 569         | KOSDAQ firms           | 2000.02–2007.07 | Lee and Kim (2009)         |
| 10.24%               | Market adjusted abnormal returns            | 91          | KOSDAQ venture firms   | 2004.01–2007.12 | Lee, Lee, and Yoon (2010)  |
| 55.53%               | Market adjusted abnormal returns            | 432         | KOSDAQ firms           | 2001.01–2005.12 | Kim and Jung (2010)        |

Note: <sup>1</sup> Initial returns are estimated as the percentage difference between the offer price and the market price on the first day of trading, and market-adjusted abnormal returns are initial returns adjusted by market returns which are the percentage difference between the market index on the offer day and the market index on the first trading day.

<sup>2</sup> The cumulative abnormal return for 20 days.

Kang (1990) reports an initial return of 99.8% and a market-adjusted abnormal-return of 97.2%. Academics in the U.S. and other countries use the terms underpricing and initial returns (not market-adjusted abnormal returns) interchangeably.

Some studies also calculate initial returns using the market prices after several days of initial trading, although they are not reported here. In Korea, the stock market has a ‘daily price limit’ of 15% restricting price fluctuations.<sup>2)</sup>

Thus, aftermarket prices may take several days before they reach the equilibrium price. In such a case, it makes sense to measure initial returns over a longer window.<sup>3)</sup>

As <Table 1> shows, the distribution of initial returns is highly skewed, with a large positive mean. There is also substantial variation in IPO underpricing over time. In the 1980s, the average initial return is 70% for 361 KSE firms, according to Jung (1992). Over the period of 1992~1996, Choi and Huh (2000) reports 69.55% for 107 KSE firms. Lee and Yi (2003) report 56.66% for 39 KSE firms over the period of 1997~2000. In the period 2000 to 2007, there were only 33 KSE firms that went public and their initial returns averaged 45.79%, according to Lee and Kim (2009), which is approximately two thirds of the average returns for the 1980s and 1990s. In contrast, over the period of 2000~2007, 569 firms reportedly went public on the KOSDAQ market with an average initial return of 58.29% accruing to investors, again Lee and Kim (2009). This drastic decrease in the number of IPOs in the KSE market inspired a large body of empirical literature in the 2000s focusing more on the IPOs occurring on the KOSDAQ market.

The degree of IPO underpricing in Korea is still large compared to other developed countries, but as <Table 1> shows, there is a substantial decrease in underpricing over time. This decreasing trend can be attributed to a learning process for each of the major parties involved: issuers, underwriters, and investors. But it is more likely that this trend is strongly associated with changes in regulations covering the mechanics and the valuation process of IPOs.<sup>4)</sup>

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2) The daily price limit was 12% prior to December 1998. Furthermore, as described in Appendix 2, there exists a change in the opening price of initial trading. Prior to June 2000, the opening price of initial trading was the offer price but now is the first asking price. These changes in the daily price limit and the opening price of initial trading could have an impact on initial returns when the aftermarket price of some IPO shares has run up to the limit.

3) Most studies presented in <Table 1> present the initial return results measured by using a variety of longer windows. Thus some studies contain many different measures for the initial return. Because of limited space, this paper presents only the initial return on the first trading day of each study.

4) Regulatory changes in the Korean IPO process, including the bookbuilding process, the allocation mechanics, and the valuation methods, are presented in Appendix 1 and 2.

Such regulatory changes could have influenced the size of initial returns. The most critical regulatory change in the pricing of IPO shares is the implementation of the bookbuilding process. Prior to bookbuilding, every IPO in Korea had to use the intrinsic pricing method to estimate the initial price of IPO shares. This practice normally sets the offer price relatively low and enables investors who successfully received an allocation to reap significant profits (see e.g., Lim (1991), Kang and Cho (2000)). Despite its significant impact of initial returns, most empirical studies have not controlled for the effect of regulatory changes on initial returns. This could lead to an invalidation of their hypotheses testing based on initial returns.

### III. Asymmetric Information Hypotheses

Among the IPO underpricing hypotheses reviewed, the hypotheses based on asymmetric information are well established and have accumulated a large body of evidence. Asymmetric information models assume that one of the major parties, such as investors, issuers, and investment bankers, knows more than the others and that such informational frictions give rise to underpricing.

#### 1. Winner's Curse

Rock (1986) assumes that investors are differentially informed such that some investors are actually better informed regarding the true value of IPO shares. Informed investors are more likely to buy underpriced shares, but other investors buy shares without distinction. Thus, other investors will be allocated more of the least desirable IPO shares, resulting in a winner's curse. Faced with this problem, the uninformed investors will purchase shares only if IPOs are underpriced sufficiently to compensate them for the bias in the allocation

of IPO shares. This requires that all IPOs should be underpriced in expectation so that the uninformed earns a non-negative expected return even after adjusted for rationing. The earliest empirical study on the winner's curse model is Kang's (1990) analysis for IPOs in the KSE market, where oversubscribed IPO shares are allocated on a pro-rata basis. Kang (1990) examines the underpricing behavior of 247 KSE firms which went public over the period of 1988~1989 and finds that the extent of underpricing is negatively related to the subscription rate which measures the likelihood of receiving oversubscribed IPOs. He also shows that average expected returns decrease significantly, from 99.8% to 2%, when adjusted for pro-rated allocations.

Since Kang's (1990) study, there has been no attempt to test the winner's curse model. But Kang's analysis on IPOs can be replicated using subscription rates reported by the literature. The unconditional expected returns reported by Choi (1999), Choi and Huh (2000), Choi (2000), Kim and Lee (2006), Lee and Joh (2007), and Lee and Kim (2009) are well above 45% during the sample period, but they decline to less than 1.1% when adjusted for subscription rates reported in their papers. Although the competition for IPO shares among investors and the subsequent rationing sharply reduces the expected returns, they never decline to less than zero. Rock's model suggests that when properly adjusted for rationing, uninformed investors earn zero expected returns and informed investors earn returns which are just enough to cover their costs of becoming informed. Since it is difficult to identify whether investors are informed or uninformed, non-zero expected returns for every investor who receives an allocation can be regarded as evidence supporting the winner's curse hypothesis.

## 2. Ex Ante Uncertainty

In line with Rock's winner's curse, Beatty and Ritter (1986) propose a hy-

pothesis that the greater is the ex ante uncertainty about the value of IPO shares, the larger is the extent of underpricing. The intuition behind this hypothesis is that being informed in the IPO market is similar to implicitly investing in a call option on an IPO.<sup>5)</sup> An increase in ex ante uncertainty will make the call option more valuable and thus induce more investors to become informed. This will further worsen the winner's curse and force issuing parties to raise the degree of underpricing to attract more uninformed investors. This explanation has a relatively simple testable implication – a positive relation between the ex ante valuation uncertainty and the degree of IPO underpricing.

All theories of underpricing based on asymmetric information share the prediction that underpricing is positively related to ex ante uncertainty, which supposedly contributes to the degree of asymmetric information. Thus, any empirical test on hypotheses based on asymmetric information must control for ex ante uncertainty. Indeed, this hypothesis has been tested by various empirical studies, but the testing has been conducted only as one of the controlling variables i.e., there has been no attempt to directly test the hypothesis.

<Table 2> summarizes the proxies for the ex ante uncertainty used in the literature, including their estimation results. As shown in <Table 2>, there is a wide range of proxies for ex ante uncertainty and their estimation results differ across studies.

<Table 2> shows that variables representing the size of IPO firms, such as total assets, sales, and gross proceeds, are the most commonly used proxies for ex ante uncertainty. However, their estimated results are not consistent across studies. It is well known that the size variables can proxy anything. A potentially more promising approach might be to identify potential uses of IPO proceeds as disclosed in the prospectus (Ljungqvist and Wilhelm, 2003) or the number of risk factors listed in the prospectus (Beatty and Welch,

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5) This option is exercised if the true value of IPO shares exceeds the offer price.

&lt;Table 2&gt; Survey of Proxies for Ex Ante Uncertainty

Evidence is collected from the academic finance journals, such as the Asia–Pacific Journal of Financial Studies, The Korean Journal of Finance (Asian Review of Financial Research), The Korean Journal of Financial Management, and The Korean Journal of Financial Studies.

| Authors                  | Proxy variables                  | Predicted sign | Results (sign)                   |
|--------------------------|----------------------------------|----------------|----------------------------------|
| Lee et al. (2010)        | Age                              | +/-            | Significant (+)                  |
|                          | Venture dummy                    | +/-            | Significant (+)                  |
|                          | Log total assets                 | +/-            | Insignificant (+)                |
| Kim and Jung (2010)      | Age                              | +/-            | Insignificant (+)                |
|                          | Gross proceeds                   | +/-            | Significant (-)                  |
|                          | Log total assets                 | +/-            | Insignificant (-)                |
| Lee and Kim (2009)       | Log age                          | +/-            | Insignificant (+/-)              |
|                          | Log gross proceeds               | -              | Insignificant (-)                |
| Lee and Joh (2007)       | Gross Proceeds                   | +              | Significant (+)                  |
|                          | Age                              | +              | Insignificant (+)                |
| Park and Shin (2007)     | Age                              | -              | Significant (-)                  |
|                          | Total assets                     | -              | Significant (-)                  |
|                          | Gross Proceeds                   | -              | Significant (-)                  |
| Kim and Lee (2006)       | Gross Proceeds                   | -              | Significant (-)                  |
|                          | Age                              | -              | Significant (-)                  |
| Park et al. (2006)       | Age                              | -              | Insignificant (-)                |
|                          | Gross proceeds                   | -              | Significant (-)                  |
|                          | Leverage                         | +/-            | Significant (-)                  |
|                          | Venture dummy                    | +/-            | Insignificant (-)                |
| Choi (2005)              | Total Assets (per share)         | +              | Significant (+)                  |
|                          | Gross Proceeds                   | -              | Significant (-)                  |
| Yoon (2003)              | Age                              | -              | Significant (-)                  |
|                          | Log total assets                 | -              | Significant (-)                  |
| Lee and Yi (2003)        | Age                              | -              | Significant (-)                  |
|                          | Log capital                      | -              | Insignificant (+)                |
| Huh et al. (2002)        | Age                              | +/-            | Significant or insignificant (-) |
|                          | Log total assets                 | +/-            | Significant (-)                  |
| Kang (2001)              | Log total assets                 | -              | Significant (-)                  |
|                          | Age                              | -              | Insignificant (-)                |
| Jang and Khil (2000)     | Venture dummy                    | +/-            | Insignificant (+)                |
|                          | Gross proceeds                   | +/-            | Insignificant (+)                |
|                          | Age                              | +/-            | Insignificant (+)                |
| Choi (1999)              | Age                              | -              | Insignificant (-)                |
|                          | Log Sales                        | -              | Significant (-)                  |
|                          | Standard deviation of net income | +              | Significant (+)                  |
| Lee, Lim, and Yon (1995) | 1/gross proceeds                 | +              | Significant (+)                  |
| Joo (1995)               | Total assets                     | -              | Significant (-)                  |
|                          | Total equity                     | +              | Insignificant (+)                |
|                          | Gross Proceeds                   | +              | Insignificant (+)                |

1996).<sup>6)</sup> These variables could provide more interesting insights on the issue.

### 3. Partial Adjustment (Information Extraction)

Bookbuilding methods give underwriters wide discretion over allocations, allowing them to elicit private information from investors which can be used in setting the offer price. In so doing, investment bankers must compensate investors through underpricing. In this framework, the ex ante measure of firm-specific uncertainty is directly related to the marginal value of investors' (positive) private information. If the firm-specific uncertainty about the value of IPO shares is great, any information that reduces such uncertainty will be valuable to an issuer, who is then expected to collect such information as much as possible via the bookbuilding process (Benveniste and Spindt, 1989), Benveniste and Wilhelm, 1990). However, when the issuer is faced with the truth-telling constraint, information gathering becomes costly in the form of underpricing as uncertainty increases. This leads to a prediction that there will be a partial adjustment of the offer price from that specified in the preliminary prospectus. More precisely, from the positive information revealed, the issuer and underwriter factor in their expectations about what the full-information price of shares will be and raise the offer price from that set prior to the bookbuilding period but set the final offer price less than the full-information price. This leads to a positive relation between the price revision over the process of bookbuilding and underpricing.

Testing this hypothesis is relatively new to the Korean IPO literature. This is because it has been just 10 years since the bookbuilding process was introduced to the Korean market. Choi (2005) has directly tested this partial-adjustment hypothesis using 390 KOSDAQ firms. His study documents the

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6) Ljungqvist and Wilhelm (2003) argue that firms intending to use their IPO proceeds to fund operating activities rather than investing activities or debt repayment are potentially more risky.

supporting evidence that underpricing is positively related to revisions in the offer price made during the bookbuilding period.<sup>7)</sup> This evidence, however, is applicable only to the sample IPO firms which adopt the comparative pricing method, such as price multiples, for the initial valuation of IPO shares. The other method commonly used for the initial pricing of IPO shares is the intrinsic pricing method based on the book value of net assets, the future profitability predicted over 2 years following the IPO year, and the market value of competing firms in the same industry. Prior to the deregulation of the pricing practice of the IPO, every IPO in Korea must adopt the intrinsic pricing method to estimate the initial price of IPO shares. This practice is known according to the literature (see e.g., Lim, 1991), Kang and Cho, 2000) that the offer price tends to be set too low and investors who successfully received allocation can reap significant gains. If the initial price of IPO shares is set too low prior to the bookbuilding process and investors who participate in the bookbuilding know it, then there will be no motivation for investors who participate in bookbuilding to reveal their private information in the partial-adjustment framework. That is, these investors can potentially benefit by not revealing positive information about new issues. For this reason, Choi (2005) argues, the partial-adjustment hypothesis is not supported when the sample IPOs use the intrinsic pricing practice.<sup>8)</sup>

On the other hand, the average initial price based on the comparative pricing practice is reportedly far less than that based on the intrinsic pricing method.<sup>9)</sup>

7) Although there has been a strict restriction on how underwriters in Korea allocate IPO shares to their clients until recently, underwriters in Korea had discretion of the allocation of IPO shares to those who participate in the bookbuilding process (high-yield funds or institutional investors). The allotment to high-yield funds and institutional investors at that time was 30% to 55% and 15% to 30%, respectively. IPO shares were allocated to high-yield funds and institutional investors participating in the bookbuilding process according to their weights determined by their credit ratings and the proximity of their bidding price to the potential offer price. Thus, the bookbuilding system at that time allowed underwriters to have discretion on the allocation to investors and provided some inducements for informed investors (high-yield funds and institutional investors participating in the bookbuilding) to reveal their information.

8) More detailed information on the current rule regarding how to set offer prices is explained in Appendix 2.

9) For instance, the average initial prices based on the comparative pricing practice and the intrinsic pricing practice are ₩8,777 and ₩10,195, respectively, according to the results presented by Choi (2005).

This would mean additional pressure on underwriters for aggressive road show activities in order to secure more (positive) information from investors. Since investors are assured of positive returns as a compensation for revealing their private information, this implies that there will be a positive relation between the price revision and underpricing for the sample IPO firms with the comparative pricing method. The author finds the evidence consistent with this prediction.

#### 4. Information Spillover

In conjunction with the partial-adjustment hypothesis, there may exist the effect of information spillover in the IPO market. If underwriters and investors deal with each other repeatedly in the IPO market, the cost of information acquisition can be reduced. Investment banks that are more active in the IPO market have a natural advantage in pricing IPOs i.e., their larger IPO deals allow them to obtain investors' cooperation more cheaply than less active underwriters can. Thus, repeated interaction creates information spillover from the primary market to the secondary market and results in less underpricing. This information spillover effect can be intensified by bundling offerings across time.<sup>10)</sup> Obtaining information from other offerings allows investors to evaluate other offerings more cheaply. Such economies of scale could result in much less underpricing. In contrast, the first firm going public (i.e., pioneering IPO firm) should compensate investors for their whole information production effort, thereby resulting in larger underpricing.

Choi and Jun (2006) have examined the effects of information spillover, bundling, and pioneering IPOs on underpricing in the Korean market. Their results indicate that the information spillover from bundling indeed decreases the marketing and information production expenditures through higher offer prices and lower underpricing. Their study also finds that pioneering IPO

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10) Refer to Ljungqvist (2006) for further detail on the information spillover hypothesis.

firms tend to have lower offer prices and larger underpricing than their followers do.

## 5. Certification Effect

The repeated interaction between investment bankers and investors as mentioned above can also create a certification effect on IPO underpricing. If an underwriter and investor interact repeatedly in the IPO market, the investor could become afraid of being excluded from all the current and future IPO deals being managed by that underwriter. This enables underwriters who are more active in the IPO market to have a natural advantage in collecting information from investors and pricing IPOs. Then, hiring underwriters who are more active and more prestigious can certify the quality of IPOs and reduce underpricing. This is because investment bankers who accumulate valuable reputation capital through repeated interactions will restrain themselves from underwriting low-quality issuers (Booth and Smith, 1986; Carter and Manaster, 1990; Michaley and Shaw, 1994).

The earliest empirical study on this issue is done by Kang (1991), who has examined the role of an underwriter's reputation on the underpricing of a Korean IPO. He first identifies prestigious and non-prestigious underwriters based on the equity size of underwriters and the size and number of IPOs undertaken by them during the period of 1988~1989. His study shows that average initial returns for prestigious and non-prestigious banks are 76.01% and 84.24%, respectively, and their difference is not significant according to t-test. He argues that one of the reasons for this weak result is the short history of the Korean IPO market.

This certification effect has been extended to examine the role of venture capitalists in the IPO market by several empirical studies (e.g., Lee, Lim and Choi, 1998; Jang and Khil, 2000; Huh et al., 2002; Yoon, 2003). The authors

suggest that because venture capitalists repeatedly bring firms to the public market, they can credibly stake their reputation that the firms they bring to market are not overvalued. They test the idea using samples of venture (capital)-backed IPOs and non-venture (capital)-backed IPOs. Their empirical results, however, are mixed. It appears that the results are quite sensitive to the market and period examined. Lee et al. (1998), report that there has been no significant relation between the reputation of venture capitalists and underpricing, after examining a matched set of 25 venture-backed and non-venture-backed KSE IPOs between 1987 and 1996.

Jang and Khil (2000) also confirm their finding that the underpricing of venture-backed IPOs is not significantly less than that of non-venture-backed IPOs, after examining a matched set of 34 KOSDAQ IPOs between 1996 and 1999.

Huh, Yoon, and Lee (2002) have examined the difference of underpricing between IPOs backed by a prestigious venture capitalist (KTB) and IPOs backed by other (non-KTB) venture capitalists. They report that the prestigious venture capitalist (KTB) is associated with lower underpricing than other venture capitalists, after examining a matched set of 65 venture-backed and non-venture-backed KOSDAQ IPOs between 1999 and 2000. They argue that KTB-backed IPOs are less underpriced because the prestigious venture capitalist (KTB) has monitored the quality of the offering.

Yoon (2003), however, finds mixed evidence for the certification effect of the prestigious venture capitalists which include KTB. He has compared underpricing of IPOs backed by the prestigious venture capitalists including KTB with that of IPOs backed by other venture capitalists using both average abnormal returns of the initial trading day and average abnormal returns of the trading day at which successive daily price limits end. His results show that there is no significant difference in abnormal returns for the initial trading

day between the two groups. But when KTB is excluded from the prestigious group, the result becomes significant at the 10% level. When the abnormal returns of the trading at which successive daily price limits end are used, IPOs backed by the prestigious group including KTB are less underpriced. But when KTB is excluded from the prestigious group, there is no difference between the two groups.

Park et al. (2006), investigate, using a sample of 343 IPO firms over the period of 2001~2003, whether an IPO firm's relationship with a merchant bank has a certification effect on underpricing. They document no supporting evidence for the certification effect of merchant banks.

The results on this issue are in general mixed. This suggests that they are sensitive not only to the sample firms and period examined, but also to the methodologies used. For instance, some studies that examined venture capitalists for KOSDAQ IPOs do not clearly explain how their prestigious venture capitalists are defined. They simply classify prestigious and non-prestigious groups based on the government regulations that guide how venture capitalists are formed. But it would be more plausible to use a proxy for the reputation capital of venture capitalists, such as the market share of the venture capital industry.

There may be an endogeneity bias on the certification effect on underpricing. An issuer doesn't select an underwriter randomly, nor does an investment bank randomly agree on which firms are to go public (Habib and Ljungqvist, 2001). The choices we actually observe are presumably made by agents. But issuers also likely base their choices on the underpricing they expect to suffer. This leads to an endogeneity bias when initial returns are regressed for underwriter choice.

The certification effect may also be impeded by a conflict of interest arising from the relationship between the underwriting business and the asset manage-

ment business which coexist in a security firm. Park and Shin (2007) examine IPOs from 1999 through 2006 and find that the extent of underpricing is larger when IPOs are underwritten by securities firms that have asset management businesses. They also find that the degree of underpricing is positively associated with the market share of the asset management business. Their findings imply that securities firms may have favored customers of their asset management business at the expense of those of the underwriting business because they can get more profit from the former which is growing much faster than the latter.<sup>11)</sup>

This does not mean that security firms make the underwriting business a scapegoat for the growth of asset management business. On the contrary, security firms expand their underwriting businesses to continuously favor customers of their asset management businesses. In so doing, they must make their underwriting business attractive and prestigious enough to attract more IPO customers, but at the same time they must deeply underprice IPO shares to satisfy the greed of their asset management customers. This conflict of interest potentially reduces the certification effect arising from the good reputation and prestige of underwriters.

#### IV. Price Stabilization

Several studies turn to the practice of price stabilization for explaining IPO underpricing. One of the services that underwriters provide in connection with an IPO is price support. This service is intended to reduce price drops in the aftermarket for a few weeks or months and used to be legally mandatory

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11) However, it is not evident that the conflict of interest occurs in the market. To be evident, we need the allocation data to examine whether the asset management business benefits from the allocation of IPO shares. Furthermore, their approach ignores the endogeneity bias mentioned above. That is, issuers also likely base their choices of underwriters on the underpricing they expect to suffer.

in the Korean IPO market. Ruud (1993) argues that the positive initial returns observed are not the result of deliberate underpricing, but the result of the price stabilization practice of IPOs whose prices threaten to fall below the offer price. The price support of underwriters reduces IPOs with negative initial returns that would be observed in the aftermarket. This censoring of the negative tail of the distribution produces positive mean initial returns even if IPOs are priced at their intrinsic value. Accordingly, Ruud hypothesizes that if there is no deliberate underpricing, then the initial return distribution of unsupported offerings should have a mean zero. This hypothesis is developed under the efficient market hypothesis.

Unlike the U.S. market where direct evidence of price support is limited because price support activities are not generally public information, underwriters in Korea must make a public announcement with regard to which IPOs are initially supported, how the intensity of stabilization varies overtime, and at what time support is withdrawn. This makes it possible to test directly the price support hypothesis in Korea.<sup>12)</sup>

Jang and Woo (1997) examine the price stabilization hypothesis proposed by Ruud. To investigate whether the observed initial returns are the byproduct of price stabilization, as Ruud proposes, or deliberate underpricing, Jang and Woo estimate average initial returns for the two hypothesized distributions of supported and unsupported IPOs. If Ruud is correct in saying that there is no deliberate underpricing, then the initial return distribution of unsupported offerings should have a mean zero. This, however, is not what they find. They find that the distribution of unsupported IPOs has a mean first-day

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12) The price stabilization hypothesis proposed by Ruud is different from the price stabilization hypothesis proposed and tested by other studies appearing in this paper such as Shin et al. (2004), Kim and Lee (2006), Lee and Kim (2009), and Lee and Joh (2007). Ruud has to investigate the price support hypothesis through the examination of the skewness of initial returns. In Korea, however, any activities supporting the aftermarket price is public information and this makes it possible to test directly the price support of underwriters. Thus, the hypothesis and results presented by those studies (except Jang and Woo, 1997) are not directly comparable to those of Ruud (1993).

return of 35.54%, while the distribution of supported IPOs has a mean first-day return of 18.86%. They argue that the mean of first-day return for unsupported IPOs is too large to justify the price stabilization hypothesis proposed by Ruud and suggest that underpricing in the Korean IPO market is caused by factors other than price support.

Kim and Lee (2006) also confirm the finding that the distribution of IPOs without the aftermarket intervention has an average 20-day cumulative abnormal return of 86.4%, while the distribution of IPOs with the aftermarket intervention has a mean 20-day cumulative abnormal return of 6%. This result also rejects the price stabilization hypothesis proposed by Ruud.<sup>13)</sup>

Shin, Chang, and Chung (2004) take a different approach to examine the effect of the aftermarket price support on underpricing. In the Korean IPO markets, price stabilization was legally mandatory until May 1999 and was abolished briefly over the period from May 1999 to February 2000. The practice, however, was reinstated from March 2000 as a recommended system rather than an obligatory system. This unique regulatory environment provides an opportunity to test their hypothesis that without the mandatory price support system, the extent of underpricing would be less because of decreased uncertainty faced by underwriters with regard to aftermarket price stabilization. They find no difference in underpricing between the two periods and argue that the reintroduction of the aftermarket price support system does not affect the magnitude of underpricing.

Their hypothesis is also examined by Kim and Lee (2006). They report that average 20-day cumulative returns are 91% before the reinstatement and 65.4% after the reinstatement and their difference is statistically significant. Their regression analysis also confirms the significant difference between them even after controlling other factors that presumably influence underpricing. Their

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13) Although Kim and Lee (2006) do not directly examine Ruud's hypothesis, they have shown the result consistent with that of Jang and Woo.

evidence thus does not support the hypothesis proposed by Shin et al. (2004), and indicates that the existence of the price support system decreases, rather than increases, the extent of underpricing.

Lee and Kim (2009), however, provide evidence that supports the price stabilization hypothesis. They analyze the data of 602 IPO stocks listed on the Korea Exchange (KRX) along with KSE and KOSDAQ from February 2000 to July 2007 and find that average initial returns are 62.26% for the period with price support and 49.17% for the period without support. Their finding is consistent with the hypothesis that the existence of price stabilization increases the extent of IPO underpricing because of the aftermarket uncertainty faced by underwriters. They also report that the effect of the non-trading period, which is defined as the period from the first day of offering to the first day of trading, is one of the important factors which determine IPO underpricing. Using the option pricing approach, they find that the option value for the non-trading period is on average 11.98% of the offering price.<sup>14)</sup> They also report that the degree of IPO underpricing increases as the option value of the IPO stock increases, and thus the loss of the option value that investors may suffer during the non-trading period is another important factor in IPO underpricing in the Korean market.<sup>15)</sup>

Price stabilization can be thought of as a put option written by underwriters and held by IPO investors, in the sense that the price support activities put a floor under initial aftermarket prices and thus act as insurance against price falls. In fact, security-issuance regulation in Korea allows investors to have the right to sell their stocks back to underwriters within the first month of

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14) They first define the non-trading period effect as the expected losses that can occur during the non-trading period and then measure an option value of the non-trading period effect with an option pricing model and analyze the effect of the option values on the IPO underpricing.

15) But, the value of an option increases as the time to expiration increases. Thus, their result may be due to the characteristics of the option pricing model. That is, the longer the non-trading period is, the larger the value of the option is. The length of the non-trading period may depend on the timing of the initial trading. Underwriters of 'cold offerings' may wait for the 'hot issue market' or the time to bundle cold offerings with hot offerings for the initial trading.

trading at 90% of the offer price.<sup>16)</sup> With such put-back options, investors investing in IPO shares are protected against price falls below 90% of the offer price, while underwriters are exposed to the risk arising from the sale of a put option. Thus the value of a put-back option reflects the risk faced by underwriters, which is similar to the risk of price stabilization.

Lee and Joh (2007) have examined the relationship between underpricing and the ex ante value of put-back options, using a sample of 174 KSE and KOSDAQ IPO firms over the period of 2003~2006. They argue that the larger the ex ante risk of the aftermarket price falling below the offer price is, the larger the ex ante value of the put-back option is. Underwriters as market makers then deliberately lower the initial offer price of an IPO to reduce the ex ante risk of the price falling below the 90% threshold. They hypothesize that the initial return and the ex ante value of a put-back option are negatively correlated and find the evidence of negative correlation between them.

However, lowering the offer price means a larger initial return (larger underpricing), rather than a smaller initial return. Contrary to their own hypothesis, this leads to a positive relation between the ex ante value of put-back option and the initial return.<sup>17)</sup> That is why they use the volatility of the industry index to which an IPO firm belongs (instead of the volatility of the market price of the IPO firm) to estimate the value of the put-back option.

This view of the origin of underpricing together with such mixed results leaves a critical question regarding the pricing mechanism of IPOs. Under the efficient market hypothesis, the market price of IPO shares reflects the true value so that positive initial returns indicate underpricing. Then any activity of the aftermarket price stabilization may implicitly indicate a failure of

16) The put-back option system existed over the period from September, 2003 to July, 2007.

17) Of course, there should be a negative relation between initial returns and the ex-post (not ex ante) value of put-back option. But what Lee and Joh (2007) attempt to find is a relation between underpricing (initial returns) and the ex ante (not ex-post) value of put-back option.

the IPO pricing mechanism. That is, the offer price is overvalued or the market price reflects bubbles or fads, instead of the true value, and thus would burst quickly leaving the market price to fall below the offer price. This suggests that a different approach may be needed to examine the relation between price stabilization and underpricing.

A possible approach is to measure underpricing using the premarket information. Kim and Lee (2006) adopt this approach and measure the extent of premarket underpricing using the ratio of the offer price to the 'intrinsic value' specified in the preliminary prospectus. This allows them to eliminate a possible introduction of fads or bubbles in measuring underpricing (Choi, 1999). Their regression result indicates that their premarket underpricing measure (the average ratio of the offer price to the 'intrinsic value') after the reinstatement is significantly larger than that before the reinstatement even after controlling other factors. This evidence is consistent with the price stabilization hypothesis: the existence of price stabilization increases the aftermarket uncertainty faced by underwriters, thereby lowering the offer price further.

## V. Hot Issue Markets

Some researchers attempt to find an explanation for underpricing from the phenomenon known as hot issue markets (e.g., Ibbotson and Jaffe, 1975; Ritter, 1984). They note that cycles exist in both the volume and average initial returns of IPOs; high initial returns tend to be followed by rising IPO volume.<sup>18)</sup> The periods of high average initial returns with rising IPO volume are referred to as 'hot issue' markets. This empirical regularity is often observed in Korea and other countries and can be hardly accounted for in the framework of

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18) One of referees points out that underpricing can exist in even cold markets.

the market efficiency. For this reason, the empirical literature turns to a behavioral explanation for the hot issue markets phenomenon.

Kim and Jung (2010) have examined 432 IPOs over the period of 2001~2005. They first identify sample IPOs based on the return of the industry index to which an IPO belongs over the 3 months preceding the closing of the order book. This market conditions variable is then split into terciles, and each IPO is assigned to one of the terciles. They find that initial returns and long-run underperformance of IPOs in the upper tercile (hot market samples) are larger than those in the lower tercile (cold market samples) and their differences are significant. Their analysis further shows that the proxies for the optimistic investors'sentiment have positive effects on initial returns and negative effects on long-term performance. Therefore, the observed initial returns of IPOs, they argue, are more likely to reflect bubbles or fads that would eventually burst rather than deliberate underpricing the favorable market conditions that prevail at the time of an offering generate over-optimism from some investors about the prospects of the offering.<sup>19)</sup>

Although Kim and Jung turn to a behavioral explanation for the phenomenon, their results provide interesting aspects to be explored further from the perspective of information acquisition. Kim and Jung's distinction of hot issue markets does not accord with the cyclical nature of hot issue markets in the literature, where they are characterized by not only high levels of initial return but also high IPO volumes (see e.g., Ibbotson and Jaffe, 1975; Ritter, 1984). This is because Kim and Jung classify the sample IPOs based on the industry-index returns rather than on the cyclical nature of IPO activities. Although their approach seems appropriate, it could ignore the effect arising from IPO volumes on underpricing. This effect is closely related to the information spillover effect from bundling IPOs in the same industry. Benveniste

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19) One of referees points out that this could result from investors'overexcitement over a certain industry at a time, as experienced during the internet bubble period.

et al. (2002), offer an alternative explanation to the hot issue markets phenomenon. They develop a model in which underwriters bundle IPOs in the same industry in order to share the costs of information production (i.e., underpricing). The central idea is that obtaining information about the industry component allows investors to evaluate other offerings from that industry more cheaply. Such economies of scale could lead to many firms in the same industry going public at the same time. This is consistent with Kim and Jung's empirical observation that companies tend to go public in industry-specific waves.<sup>20)</sup>

Kim and Jung's results also imply that initial returns can be predicted using market conditions, which are measured by industry-index returns in the 3-month period prior to the offerings, prevailing prior to the offering that is, there is a positive relation between the pre-market condition and initial returns. This contradicts the information extraction theory proposed by Benveniste and Spindt (1989), since the industry-index returns are public information which is freely available, so there is no need to compensate investors for it by leaving money on the table.

## VI. Long-Run Performance

Another unique pattern associated with IPOs is the poor aftermarket performance of IPOs in the long run. Measured from the aftermarket prices over the various time frames, <Table 3> summarizes the evidence on the long-term performance of the Korean IPOs documented by empirical studies since Lim and Lee (1995).

As shown in <Table 3>, there are in general three measures used to show

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20) Booth and Chua (1996) also argue that when there is a sequence of many offerings in the market, the marginal cost of information production is less, thereby reducing underpricing.

the long-term performance: cumulative abnormal returns (CARs), buy and hold abnormal returns (BHARs), and relative wealth (WR). Results presented in <Table 3> show a declining tendency of long-term performance of IPOs over a period of 36 months and these measures appear to be different across studies.<sup>21)</sup> This suggests that they are sensitive to the methods and benchmarks used. Thus, research on this area has been subject to more rigorous empirical testing. Fama (1970) particularly emphasizes that all models for expected returns are incomplete descriptions of the systematic patterns in average returns during any sample period. As a result, tests of performance are always contaminated by a bad-model problem. The bad-model problem is less serious in performance studies that focus on short return windows (a few days), although the problem grows with the return horizon.

To mitigate the problem, Choi and Huh (2000) have used various benchmarks that are carefully constructed to avoid known biases. They have also adopted a monthly calendar-time portfolio approach for measuring long-term abnormal performance. By forming monthly calendar-time portfolios, all cross-correlations of event-firm abnormal returns are automatically accounted for in the portfolio variance. Any methodology that ignores cross-sectional dependence of event-firm abnormal returns that are overlapping in calendar-time is likely to produce overstated test statistics (Mitchell and Stafford, 1998). Their results show that long-term abnormal returns based on a calendar-time portfolio approach are 0.31% for 12 months, 0.48% for 24 months, and 0.54% for 36 months, confirming the poor long-term performance of IPOs.<sup>22)</sup>

21) The result of Kim and Kim (2000) presented in <Table 3> is the only one that shows positive long-term performance over a 36-month period. Their long-term returns are adjusted only by the return of control firms comparable to IPO firms. Their tests of the long-term performance could be contaminated by a bad-model problem as suggested by Fama (1998). One way to mitigate the problem is to adjust returns of IPO shares by various benchmarks which are carefully constructed to avoid known biases, as suggested by Choi and Huh (2000).

22) Returns are estimated by the Fama-French three factor model. The three factors are zero-investment portfolios representing the excess return of the market; the difference between a portfolio of small stocks and big stocks, SMB; and the difference between a portfolio of high BE/ME stocks and low BE/ME stocks, HML. BE/ME is the ratio of the book value of equity to the market value of equity.

<Table 3> Survey of Long-Term Performance of Korean IPOs

*BHARs* are calculated as the difference between the average monthly returns for the event firms and the benchmark portfolios. *CARs* are calculated by summing the average monthly abnormal returns. The wealth relative (*WR*) is the average gross return of the sample firms divided by the average gross return of the benchmark.

| Author(s)                        | Sample period | Sample size | Under pricing (%) | Market | Period (months) | Performance measure |          |      |
|----------------------------------|---------------|-------------|-------------------|--------|-----------------|---------------------|----------|------|
|                                  |               |             |                   |        |                 | CAR (%)             | BHAR (%) | WR   |
| Lim and Lee <sup>1</sup> (1995)  | 1980~1990     | 331         | 46.77             | KSE    | 6               | 2.38                | -        | -    |
|                                  |               |             |                   |        | 12              | 1.33                | -        | -    |
|                                  |               |             |                   |        | 24              | -4.96               | -        | -    |
|                                  |               |             |                   |        | 36              | -11.53              | -        | -    |
| Lim (1997)                       | 1988~1994     | 342         | 75.45             | KSE    | 12              | -                   | -        | 0.99 |
|                                  |               |             |                   |        | 24              | -                   | -        | 0.96 |
|                                  |               |             |                   |        | 36              | -                   | -        | 0.95 |
| Choi and Huh <sup>5</sup> (2000) | 1992~1996     | 107         | 70.21             | KSE    | 12              | -0.31               | -        | 0.99 |
|                                  |               |             |                   |        | 24              | -0.48               | -        | 0.98 |
|                                  |               |             |                   |        | 36              | -0.54               | -        | 0.93 |
| Kim and Kim <sup>3</sup> (2000)  | 1988~1994     | 267         |                   |        | 6               | 4.47                | 2.31     | -    |
|                                  |               |             |                   |        | 12              | 5.47                | 1.04     | 1.14 |
|                                  |               |             |                   |        | 24              | 10.46               | -0.12    | 1.01 |
|                                  |               |             |                   |        | 36              | 18.87               | -3.76    | -    |
| Choi (2000) <sup>2</sup>         | 1992~1998     | 141         | 114.37            | KSE    | 36              | 7.71                | 2.94     | -    |
| Lee and Yi <sup>1</sup> (2003)   | 1997~2000     | 39          | 56.66             | KSE    | 6               | 9.27                | -        | 1.63 |
|                                  |               |             |                   |        | 12              | -6.6                | -        | 1.56 |
|                                  |               |             |                   |        | 24              | 10.02               | -        | 1.93 |
|                                  |               |             |                   |        | 36              | -21.31              | -        | 1.80 |
|                                  |               | 131         | 14.98             | KOSDAQ | 6               | 25.58               | -        | 1.53 |
|                                  |               |             |                   |        | 12              | 31.9                | -        | 1.30 |
|                                  |               |             |                   |        | 24              | 14.01               | -        | 1.35 |
|                                  |               |             |                   |        | 36              | -7.83               | -        | 1.08 |
| Kim and Jung <sup>4</sup> (2010) | 2001~2005     | 432         | 55.53             | KOSDAQ | 6               | -5.25               | -13.71   | -    |
|                                  |               |             |                   |        | 12              | -11.34              | -23.31   | -    |
|                                  |               |             |                   |        | 24              | -25.72              | -50.53   | -    |
|                                  |               |             |                   |        | 36              | -30.81              | -90.67   | -    |

Note : <sup>1</sup> Returns (CAR) are adjusted to control for the market risk premium.

<sup>2</sup> Returns (CAR & BHAR) are adjusted to control for the benchmark return based on size portfolios.

<sup>3</sup> Returns (CAR & BHAR) are adjusted to control for the return of control firms.

<sup>4</sup> Returns (CAR & BHAR) are adjusted to control for the return of Fama-French portfolios.

<sup>5</sup> Returns are estimated by the Fama-French three factor model. The three factors are zero-investment portfolios representing the excess return of the market; the difference between a portfolio of small stocks and big stocks, SMB; and the difference between a portfolio of high BE/ME stocks and low BE/ME stocks, HML. BE/ME is the ratio of the book value of equity to the market value of equity.

The large initial returns together with the poor long-term performance, as shown in <Table 3>, cast doubt on the explanations based on information asymmetry and price stabilization. Some researchers, thus, turn their attention to behavioral explanations for IPO underpricing. Behavioral theories assume that the favorable market conditions that prevail at the time of an offering generate over-optimism from some investors about the prospects of IPO firms. Such investors bid up the price of IPO shares beyond true value and the aftermarket share price eventually falls below the IPO price in the months following the offering. There are several empirical studies that build on this explanation.

Choi (2000) has examined the effect of optimistic investors' sentiment on the large initial returns and the poor aftermarket performance. He uses the subscription rate at the time of offering as a proxy for investors' demand for IPOs at the time of trading and reports its positive relation with the initial return but negative relation with long-term performance.<sup>23)</sup>

Kim and Jung (2010) also examine the relation between investors' sentiment and the underpricing phenomenon. They report that the proxies for the optimistic investors' sentiment, as measured by the volatility of initial returns, have positive effects on initial returns and negative effects on long-term performance. The result of these studies suggests that the observed initial returns of IPOs are more likely to reflect bubbles or fads rather than deliberate underpricing.

Several studies attempt to relate long-term underperformance to poor operating performance occurring since the offering. If managers attempt to window-dress their accounting numbers prior to going public, this will lead to pre-IPO performance being overstated and post-IPO performance being understated. Managers of IPO firms also tend to offer their issues to coincide

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23) Benveniste and Spindt's (1989) model also suggests that underpricing should be concentrated among the IPOs drawing the highest level of pre-market interest.

with periods of unusually good performance levels, which cannot be sustained in the future.

Lim (1997) find that firms going public exhibit a substantial decline in ROA, ROE, and total assets turnover ratio over the 4 years from the offering. Kim and Kim (2000) also confirm Lim's result that post-issue sales growth, account receivables turnover ratio, and ROS decline significantly over the 3 years after the IPO, as compared to those ratios in the IPO year.

Choi and Kim (2008) examine underwriters' opportunistic earnings forecast behavior and its relation with the long-term performance of IPO firms. They find that the earnings forecast errors of IPO firms, which are estimated by using earnings forecasts made at the time of filing the registration statements, are positive and significantly large, suggesting that their future earnings have been too optimistic. They also report that such forecast errors have positive effects on initial returns and negative effects on long-term performance.

Overall, these studies report that the performance of IPO firms declines significantly relative to their pre-IPO levels, based on several performance measures. Further, they present evidence to support the contention that high pre-IPO operating performance levels may lead investors to develop optimistic assessments of earnings growth for the IPO firms.

The natural question that arises is whether the long-run decline in operating performance subsequent to the IPO is anticipated by the market. In the efficient market framework, the market is not fooled by this behavior and correctly anticipates and accounts for it in its valuation of the firm. If the market is able to account for window dressing, the long-term stock price performance of IPOs should be normal. As we have seen in the literature, however, the decline in operating performance is not anticipated and investors are constantly surprised by the poor operating performance of IPO firms, resulting in the long-term underperformance of IPO shares. Therefore, the results in the liter-

ature suggest that potential investors initially have high expectations of future earnings growth, which are not subsequently fulfilled, and bid up the after-market price of IPO shares beyond true value at the time of initial trading. This is also consistent with behavioral explanations.

## VII. Concluding Remarks

This paper reviews the hypotheses and empirical evidence that have been proposed to explain IPO activities and pricing behavior in the Korean market. In particular, the paper reviews empirical evidence and explanations related to the three patterns associated with IPOs: underpricing(positive initial returns), hot issue markets, and long-run underperformance.

A number of hypotheses have been offered to explain large initial returns in the Korean IPO market, focusing on various aspects of the relations between underwriters, issuers, and investors. In general, these hypotheses can be grouped under two broad categories: deliberate underpricing and behavioral approaches. The first category can be further divided into two subcategories: asymmetric information and price stabilization. While the first subcategory attempts to explain underpricing in the framework of the efficient market and consider it as a part of necessary costs associated with initial public offerings, the last subcategory turns to one of services that underwriters provide in connection with IPOs - the price stabilization practice which is intended to reduce price drops in the aftermarket. Behavioral approaches regard initial returns as fads or bubbles caused by irrational investors who bid up the price of IPO shares beyond true value.

Among the hypotheses of IPO underpricing reviewed, the hypotheses based on asymmetric information are well established and supported by the empirical literature. However, the significant variation in the degree of underpricing over

time (e.g., hot and cold issue markets) and the poor long-term performance make many researchers doubtful whether information-based theories could be enough to explain large initial returns. For this reason, some researchers turn their attention to behavioral explanations which put more emphasis on the effect of investors' irrational behavior or sentiment on stock prices.

The hot issue market phenomenon and the long-term underperformance of IPOs suggest that if the market conditions prevailing at the time of an offering are favorable, optimistic assessments can be generated from investors about the prospects of the offering, thereby resulting in abnormalities in the aftermarket price. This evidence is not consistent with the efficient market and calls into question the explanatory power of the information-based hypotheses. Abnormalities in the market price can also shed some light on the mixed evidence presented by the empirical studies reviewed in this paper, particularly studies related to the price stabilization and the certification effect. A different approach may be needed to examine the relation between the information-based hypotheses and underpricing. One of potentially promising approaches is to measure underpricing using the premarket information. This will allow us to eliminate a possible introduction of abnormalities, such as fads or bubbles, in measuring underpricing.

Behavioral approaches to IPO underpricing are still in the early stage of development and one of the potentially promising research areas. There have been several studies that examine the effect of investors' irrational behavior or sentiment on the aftermarket price of IPOs but no studies on the effect of the behavioral biases of issuers and underwriters on underpricing.

Evidence documented in the literature suggests that there is no dominant explanation for the large initial returns observed in the Korean IPO market. Thus, the hypotheses reviewed here are not mutually exclusive and a given hypothesis can be more relevant to some IPOs than to others.

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## <Appendix 1> Regulatory Changes in the Bookbuilding and Allocation Process of Korean IPOs

| Year and Month | Bookbuilding process and allocation of IPO Shares                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1997. 8        | <ul style="list-style-type: none"> <li>• Implement the bookbuilding system.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2000. 2        | <ul style="list-style-type: none"> <li>• The standardized bookbuilding process is proposed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2000. 6        | <ul style="list-style-type: none"> <li>• A band of the desired offer price should be proposed to investors who participate in the bookbuilding.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2000. 12       | <ul style="list-style-type: none"> <li>• Abolish the standardized bookbuilding process.</li> <li>• Underwriting rules are enacted by Korea Financial Investment Association.</li> <li>• Allocate IPO shares proportionately to ESOP (20% for KSE, 0~20% for KOSDAQ), regular investors (20% for KSE, 15~35% for KOSDAQ), institutional investors (15% for KSE, 10% for KOSDAQ), and high-yield funds (45% for KSE, 55% for KOSDAQ).</li> <li>• Allocate IPO shares to institutional investors who submit a bidding price within the range from top 15% to bottom 15% of the bidding prices.</li> <li>• Allocate IPO shares to institutional investors participating in the bookbuilding according to their weight determined by their credit ratings and the proximity of their bidding price to the offer price.</li> </ul> |
| 2002. 7 (8)    | <ul style="list-style-type: none"> <li>• The method and process of the bookbuilding are liberalized and autonomously determined by underwriters and issuers.</li> <li>• Implement the over-allotment option or Green Shoe option which allows underwriters to borrow outstanding shares (up to 15% of IPO shares) at the offer price from large shareholders and allocate them to investors on the pro-rata basis.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                |
| 2003. 9        | <ul style="list-style-type: none"> <li>• The allocation of IPO shares to high-yield funds is gradually reduced by Sept. 2004 from 45% (KSE) and 55% (KOSDAQ) to 30% (KSE and KOSDAQ). Remaining shares are allocated to institutional investors.</li> <li>• Revise the over-allotment option to allow underwriters to borrow shares at the 95% of the offer price from large shareholders.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2005. 4 (7)    | <ul style="list-style-type: none"> <li>• Remove the mandatory 30% allocation to high-yield funds. Allocation to institutional investors becomes 60% of IPO shares.</li> <li>• Revise the over-allotment option to borrow shares at the 90% of the offer price.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2007. 7        | <ul style="list-style-type: none"> <li>• A proposal to advance the underwriting process of IPOs is proposed.</li> <li>• Reduce a period between the bookbuilding and the initial trading from two weeks to one week.</li> <li>• IPOs in the KSE market can be offered with existing shares as well as new issues. For the KOSDAQ IPOs, only new issues are allowed.</li> <li>• Revise the over-allotment option to borrow shares at the 100% of the offer price from large shareholders and allocate them autonomously to investors.</li> </ul>                                                                                                                                                                                                                                                                              |

## <Appendix 2> Regulatory Changes in the Price Support and the Pricing Methods of Korean IPOs

| Year and Month | Price support                                                                                                                                                                                                                    | Pricing methods                                                                                                                                                                                                                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1991. 6        |                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Determine the offer price within the arithmetic mean of the 'intrinsic value' and the 'relative value' of comparable firms.</li> <li>• The opening price of initial trading is the offer price.</li> </ul>       |
| 1996.8         |                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Determine the offer price through an agreement between underwriters and issuing firms based on the intrinsic and relative values, and market situation, etc.</li> </ul>                                          |
| 1997. 8        | <ul style="list-style-type: none"> <li>• Abolish the mandatory price support system</li> </ul>                                                                                                                                   |                                                                                                                                                                                                                                                           |
| 1998.11        | <ul style="list-style-type: none"> <li>• The mandatory price support is re-instated to protect IPO investors against price falls below 90% of the offer price for one month after the initial trading.</li> </ul>                |                                                                                                                                                                                                                                                           |
| 1999.5         |                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Determine the offer price based on the weighted average price established through the bookbuilding process.</li> </ul>                                                                                           |
| 2000. 2        | <ul style="list-style-type: none"> <li>• Maintain the aftermarket price above the 80% of the offer price for (at least) 1 month after the initial trading.</li> <li>• At least 50% of IPO shares should be supported.</li> </ul> |                                                                                                                                                                                                                                                           |
| 2000. 6        | <ul style="list-style-type: none"> <li>• Maintain the aftermarket price above the 80% of the offer price for 2 months after the initial trading.</li> <li>• 100% of IPO shares should be supported.</li> </ul>                   | <ul style="list-style-type: none"> <li>• Determine the offer price within <math>\pm 10\%</math> of the weighted average price established through the bookbuilding.</li> <li>• The opening price of initial trading is the first asking price.</li> </ul> |
| 2000. 12       | <ul style="list-style-type: none"> <li>• The aftermarket price is supported for a period over 1-6 months after the initial trading.</li> </ul>                                                                                   | <ul style="list-style-type: none"> <li>• Determine the offer price within <math>\pm 30\%</math> of the weighted average price established through the bookbuilding.</li> </ul>                                                                            |

| Year and Month | Price support                                                                                                                                                                                                                                                     | Pricing methods                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2002. 7 (8)    | <ul style="list-style-type: none"> <li>• Maintain the aftermarket price above the 90% of the offer price for (at least) 1 month after the initial trading.</li> <li>• 100% of IPO shares should be supported.</li> </ul>                                          | <ul style="list-style-type: none"> <li>• The offer price is autonomously determined by underwriters and issuers based on the bookbuilding.</li> <li>• Remove from the underwriting regulation the details of the pricing of IPO shares such as the method of intrinsic and relative valuation.</li> <li>• Determine the offer price within <math>\pm 10\%</math> of the weighted average price established through the bookbuilding process.</li> </ul> |
| 2003. 9        | <ul style="list-style-type: none"> <li>• Abolish the mandatory price support system.</li> <li>• Introduce the put-back option system that provides investors with a right to sell IPO shares back to underwriters at more than 90% of the offer price.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2007. 7        | <ul style="list-style-type: none"> <li>• Abolish the put-back option system.</li> </ul>                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• The top tier price offered in the bookbuilding process can be considered in determining the final offer price.</li> </ul>                                                                                                                                                                                                                                                                                      |